E1 244 - Detection and Estimation Theory (Jan 2026)

Assignment 6

Submission Date: 1 April, 2026
Total Marks: 50

General Assignment Guidelines:

- *Submission via Teams:* Assignments will be allocated to you via the Teams "Assignments" feature. You will have to upload your answer sheet via the same feature in Teams. **Answer sheets sent to me or the TAs by e-mail will not be considered.** Please familiarize yourself with this feature.
- *Late Submission Policy:* A delay of 0-24 hours beyond the deadline will result in 25% penalty. Assignments cannot be submitted beyond the delay of 24 hours.

  The upload time reflected in the Teams will be considered as final. You are highly encouraged to upload your answer sheets well before the deadline so that any potential connection/technical issues can be resolved in time (recommended time is at least 1 hour before the deadline).
- *File Logistics:* You can scan your handwritten answers, or use a tablet, or typeset your answers in Latex. Following logistics should be followed while uploading the answer sheet:
  - Make sure your scans are properly visible. You can try using some scan apps (Ex. AdobeScan) to get better results.
  - The total file size should not be too large (ideally, less than 5 Mb). You can use available apps/softwares to reduce the file size. Make sure that the answers are clearly visible while reducing the size.
  - Only a single pdf file is allowed to be uploaded (other formats will not be accepted).
  - The order of answers in your file should be in the same as the order of questions. Check that you have included all the pages in your file before uploading it.
  - Name you file as: DET_Assignmentx_FirstName, where x is the assignment number (x=1,2,...)
  - Mention your name, course name and submission date on the first page.
- *Collaboration Policy:* You are allowed to discuss the concepts/questions with your classmates. However, the final answer should be **as per your own understanding**. Merely copying solutions from classmates/online sources will attract significant penalty and strict disciplinary action (refer to Section 13.2 and 13.3 of the IISc student information handbook - link). If you have collaborated/discussed with other classmates or referred some online resource for a particular question, **clearly mention the names of the classmates/online resource at the beginning of your answer.**

All problems carry 10 marks but may not be equally difficult.

Problem 1 A bullet is fired at at one of the two coordinate locations - $H_0 : (x_0, y_0, z_0)$ or $H_1 : (x_1, y_1, z_1)$. The bullet does not necessarily hit the chosen target precisely. **In each coordinate**, the distance of the of the bullet landing location and the actual target location is a Gaussian R.V. with zero mean and variance σ^2. The R.V.s for all coordinates are independent. We wish to determine which of the two hypothesis was correct based on observing the location of the fired bullet.
(a) Formulate the binary hypotheses testing problem and compute the likelihood ratio.

(b) Derive the expression for P_F and P_D for any general threshold γ. Will the ROC curve in this problem look like the ROC curve for the "DC signal in noise" example covered in the class? If yes, what is the expression of d here (in the class example, it was $\frac{a}{\sigma}$)?

(c) Use any software/language to draw the ROC curve for different values of d.

Problem 2 Let X and Y be two random variables whose density is given in the below figure. The density is constant c in the shaded region and is zero in the non-shaded region. Let H_0 be the hypothesis that $X \leq 0$ and H_1 be the hypothesis that $X > 0$. The random variable Y is observed.
(a) Find the value of the constant c. Find $\mathrm{P}(H_0)$ and $\mathrm{P}(H_1)$.

(b) Derive the expression of $f(y/H_0)$ and $f(y/H_1)$. Clearly sketch these densities in two separate figures.

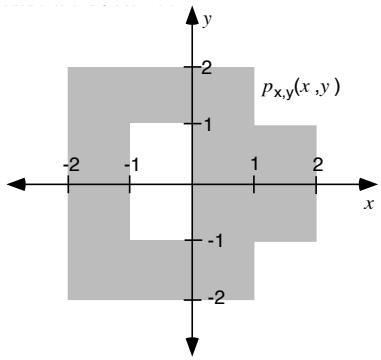

Figure 1: Joint density $f_{X,Y}(x, y)$

(c) Derive a decision rule that decides between H_0 and H_1 based on observing $Y = y$ and minimizes the total error probability. Find the decision regions Z_0 and Z_1 corresponding to this decision rule. Sketch these regions.

(d) Derive the error probability for the above decision rule.

Problem 3 Let the components of the measurement vector $\mathbf{y} = [y_1, \ldots, y_d]^T$ be binary valued (0 or 1). Consider a M-ary testing problem for some hypotheses $H_j, j = 0, 1, \ldots, M-1$. Let $P(H_j)$ be the prior probability of hypothesis H_j. Define

$$p_{ij} = \Pr\{y_i = 1 \mid H_j\} \quad \forall \quad i = 1, 2, \cdots, d, \text{ and } j = 0, 1, \cdots, M-1$$

with the components of $\mathbf{y}$ being statistically independent given hypothesis H_j.
(a) Show that the minimum probability of error is achieved by the following decision rule: Decide H_j if $f_j(\mathbf{y}) \geq f_k(\mathbf{y})$ for all $k \neq j$, where

$$f_j(\mathbf{y}) = \sum_{i=1}^{d} y_i \ln \frac{p_{ij}}{1 - p_{ij}} + \sum_{i=1}^{d} \ln(1 - p_{ij}) + \ln P(H_j).$$

(b) Let $M = 2$, d be odd, $P(H_0) = P(H_1) = 0.5$, and

$$p_{i0} = p < 1/2, \quad p_{i1} = 1 - p, \quad \forall \quad i = 1, \ldots, d$$

Show that the minimum error rate decision rule becomes

$$\text{Decide } H_1 \text{ if } \sum_{i=1}^{d} y_i > d/2, \text{ and } H_0 \text{ otherwise.}$$

(c) Show that the probability of error of the above rule in part (b) is given by

$$P_e = \sum_{k=0}^{(d-1)/2} \binom{d}{k} p^{d-k} (1-p)^k.$$

Problem 4 Consider the following M-ary testing problem: $H_k : Y \sim \text{Poisson}((k+1)\lambda), \quad k = 0, 1, \cdots, M-1$. Assume uniform costs and equal priors.

(a) Find the optimal Bayes test.

(b) Find an expression for the boundaries of the decision region.

(c) Indicate how would you compute the probability of error.

Problem 5 Consider the following testing problem: $H_0 : f(y/H_0) = \frac{1}{2}e^{-|y|}$ versus $H_1 : f(y/H_1) = e^{-2|y|}$, where $y \in \mathbb{R}$. Consider the following three tests: $\delta_1(y) = 0$ for all y, $\delta_2(y) = 1$ for all y. The third test $\delta_3(y)$

is a randomized test given by:

$$\delta_3(y) = \begin{cases} 1 \text{ with probability } 0.5 & \text{if } y < 0, \\ 1 & \text{if } 0 \leq y \leq \tau, \\ 0 & \text{if } y > \tau, \end{cases}$$

where $\tau \geq 0$ is a given constant.

(a) Compute and sketch the ROC for test δ_3.

(b) Show that a suitable randomization between tests δ_0, δ_1, and δ_3 improves the ROC curve for $P_F = [0, \alpha_0] \cup [\alpha_1, 1]$, where $\alpha_0 = \frac{\sqrt{3}}{4}$ and $\alpha_1 = 1 - \alpha_0$.

(c) Show that the corresponding ROC for part (b) is a straight line for $P_F = [0, \alpha_0]$, coincides with the ROC of δ_3 for $P_F = [\alpha_0, \alpha_1]$ and is another straight line for $P_F = [\alpha_1, 1]$.